

ID	Category	Sub-Type
31	32 Access Control	Ownable
32	33 Access Control	RBAC
33	34 Access Control	Timelock
34	35 Access Control	Guardian
35	36 Access Control	MultiSig
36	37 Access Control	Claimable
37	38 Access Control	Permit Roles
38	39 Access Control	Revocation
138	139 Compliance Controls	Blacklist
139	140 Compliance Controls	Whitelist
140	141 Compliance Controls	Sanctions Oracle
141	142 Compliance Controls	KYC NFT
113	114 Cross-Chain & Bridging	Lock & Mint
114	115 Cross-Chain & Bridging	Prove & Claim
115	116 Cross-Chain & Bridging	Timeout
116	117 Cross-Chain & Bridging	Replay
117	118 Cross-Chain & Bridging	Gas Escrow
118	119 Cross-Chain & Bridging	Finality Window
132	133 DoS & Gas	Unbounded Loop
133	134 DoS & Gas	Griefing via Revert
134	135 DoS & Gas	Block Gas Limit
135	136 DoS & Gas	Storage Blowup
136	137 DoS & Gas	Event Flood
137	138 DoS & Gas	Pull Queue
55	56 ERC-20 Conformance	Transfer
56	57 ERC-20 Conformance	Approve
57	58 ERC-20 Conformance	TransferFrom
58	59 ERC-20 Conformance	Permit
59	60 ERC-20 Conformance	Zero Address
60	61 ERC-20 Conformance	Overflow/Underflow
61	62 ERC-20 Conformance	Mint/Burn Access
62	63 ERC-20 Conformance	Decimals/Name/Symbol
63	64 ERC-20 Conformance	Events
64	65 ERC-20 Conformance	Blacklist/Pause
83	84 Economic & MEV Safety	Slippage MinOut
84	85 Economic & MEV Safety	Price Band
85	86 Economic & MEV Safety	Sandwich Defense
86	87 Economic & MEV Safety	Flash Loan Check
87	88 Economic & MEV Safety	Cooldown
88	89 Economic & MEV Safety	Velocity Caps
89	90 Economic & MEV Safety	Max Position
23	24 Events & Logs	Emission Coverage
24	25 Events & Logs	Ordering
25	26 Events & Logs	No Ghost Events
26	27 Events & Logs	Indexing
27	28 Events & Logs	Reorg Handling
28	29 Events & Logs	Overflow Topics
29	30 Events & Logs	Sparse Ranges
30	31 Events & Logs	ABI Decode Fail
107	108 Governance	Delegate
108	109 Governance	Proposal Flow
109	110 Governance	Quorum

110	111 Governance	Cancel
111	112 Governance	Upgrade via Gov
112	113 Governance	Attack Sim
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2	3 Interface & ABI	Selector Mapping
3	4 Interface & ABI	Error Selector
4	5 Interface & ABI	Event Topics
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120	121 Math & Accounting	Overflow
121	122 Math & Accounting	Fee Accounting
122	123 Math & Accounting	Dust Handling
123	124 Math & Accounting	Invariant Sums
124	125 Math & Accounting	Precision Loss
125	126 Math & Accounting	Signed Ints
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98	99 Meta-Transactions & Account Abstraction	AA Sig
99	100 Meta-Transactions & Account Abstraction	AA PostOp
100	101 Meta-Transactions & Account Abstraction	Intent Packing
65	66 NFT Standards	Mint721
66	67 NFT Standards	Transfer721
67	68 NFT Standards	Approval721
68	69 NFT Standards	URI721
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72	73 NFT Standards	Enumerable
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76	77 Oracle & Market Data	Deviation
77	78 Oracle & Market Data	Quorum
78	79 Oracle & Market Data	Fallback TWAP
79	80 Oracle & Market Data	Sequencer Uptime
80	81 Oracle & Market Data	Scaling
81	82 Oracle & Market Data	Freeze
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44	45 Pausability & Circuit Breakers	Circuit Trip
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20	21 State-Changing Writes	Permit+Action
21	22 State-Changing Writes	Nonce Gap
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51	52 Upgradeability / Proxy	Delegatecall Attack
52	53 Upgradeability / Proxy	EIP-1967 Slots
53	54 Upgradeability / Proxy	Upgrade with Data
54	55 Upgradeability / Proxy	Beacon Proxy

Test Name	Interaction	What to Do
owner-only functions	write	Call as non-owner then owner
roles & admin roles	write	Grant/revoke roles, test guards
queued operations	write	Queue->wait->execute
emergency stop	write	Trigger guardian function
threshold signing	offchain	Propose->sign->execute
two-step ownership	write	transferOwnership->accept
sig-based role grant	write	EIP-712 role grant
role/token revocation	write	Revoke & test access
blocked addresses	write	blacklist then transfer
allowed only	write	non-whitelisted call
external check	read	mock oracle says flagged
soulbound gating	write	no-KYC user
source deposit	cross-chain	approve+lock()
target chain	cross-chain	submit proof
retry/timeout	cross-chain	expire message
message idempotency	cross-chain	resubmit proof
fee on target	cross-chain	insufficient fee
optimistic rollup	cross-chain	withdraw -> finalize
large arrays	write	process many items
external call revert	write	callee reverts
upper bound	write	near block limit
mapping growth	write	fill large mapping
log spam	event	emit many
withdraw queue depth	write	large queue
happy-path	write	transfer()
set allowance	write	approve()
spend allowance	write	transferFrom()
EIP-2612	write	permit() then transferFrom()
rejections	write	transfer to 0
safe math	write	edge values
role-guarded	write	mint/burn
metadata reads	read	view calls
Transfer/Approval	event	run ops
compliance	write	blacklist+transfer
DEX swap	write	swap with minOut
bounds	write	order outside band
commit-reveal	write	commit->reveal
no-fee drain	write	flashloan attempt
withdrawal delay	write	attempt early
per-addr TPS	write	flood tx
position limits	write	exceed limits
All state writes emit events	event	Exercise all paths
Event order per operation	event	Trigger multiple emits
Reads don't emit	event	Call views
Filters by topic	event	Query getLogs with topics
Remove/add logs on reorg	event	Simulate reorg on fork
Many events in block	event	Emit 1k events
Wide block scan	event	Scan large range
Malformed log data	event	Fuzz topic/data length
voting power	write	delegate()
propose->queue->execute	write	full lifecycle
quorum reached	read	getVotes at snapshot

authorized cancel	write	cancel()
gov controls proxy	write	execute upgrade
flash-loan voting	write	attempt with snapshot rules
Decode view return types	read	eth_call on multiple view fns
Encode complex inputs	read	Staticcall with encoded args (structs, arrays)
Function selector collisions	read	Compare selectors; call via selector
Custom errors ABI	write	Trigger revert paths
Event signature & indexed topics	event	Emit events; capture logs
down/up rules	read	calc with edge values
mul/div add/sub	write	max uint ops
fee skim	write	apply fee
tiny amounts	write	transfer dust
conservation	write	trade/settle
decimal scaling	read	scale repeated
negatives	write	borrow/repay negatives
perps math	read	calc funding
trusted forwarder	write	execute via forwarder
UserOp nonce	write	resubmit same op
gas sponsorship rules	write	invalid paymaster sig
aggregated signature	write	invalid agg
refund/accounting	write	out-of-gas in postOp
callData packing	write	malformed callData
safe mint	write	safeMint()
safe transfer	write	safeTransferFrom()
approve operator	write	setApprovalForAll()
tokenURI	read	read URI
batch mint	write	mintBatch()
safe batch	write	safeBatchTransferFrom()
ERC2981	read	royaltyInfo()
ERC721Enumerable	read	totalSupply()
metadata flow	write	reveal()
permit-like	write	permit for 721 if supported
emission ratio	event	trace all writes
custom errors	write	trigger each error
ABI docs	offchain	docgen
if enabled	offchain	toggle flags
staleness	read	read latestRoundData()
drift threshold	read	compare prev vs new
multi-agg	read	aggregate N sources
AMM twap	read	uniV2 twap calc
L2 flag	read	read uptime feed
decimals normalization	read	normalize to 18
halted prices	read	frozen feed
insane values	read	fuzz large values
paused blocks writes	write	pause(); attempt writes
resume operations	write	unpause(); retry
who can pause	write	non-pauser attempt
pause subset	write	pause(feature)
velocity caps	write	rapid calls
error budget burn	write	Inject repeated failures
Simple getter returns	read	Call views
Multicall aggregation	read	Aggregate via Multicall
blockTag at N-100, hash	read	Call at historical block

eth_getStorageAt / mapping slot math	read	Compute slot & fetch
Empty/zero state	read	Zero/None cases
onlyRole-view guard	read	Role-limited view
eth_call mirrors state	read	Simulate state-changing via staticcall
Confirm state across blocks	read	Compare consecutive heads
checks-effects-interactions	write	call external then internal
nonReentrant	write	recursive call
withdraw pattern	write	multiple withdrawals
hooks	write	send with hook
A->B->A path	write	craft path
chainId mismatch	offchain	sign on chain A, send on B
increment per owner	write	reuse permit
expired	write	past deadline
allowance bounds	write	spend > allowance
v/r/s bounds	offchain	low-s / zero-v
zero addr protection	offchain	ecrecover(0)
Happy-path state write	write	Call core mutating fn
Multiple writes in 1 tx	write	Use on-chain Batch/Multicall
Re-submit tx replacement	write	Same nonce higher gas
ChainId & nonce	write	Try replay on fork with diff chainId
EIP-2930	write	Send with access list
Error paths	write	Trigger each require/revert
EIP-1559 bounds	write	maxFee < basefee -> expect revert
Approve+use in same tx	write	Call permit then transferFrom
Multiple pending tx	write	Send out-of-order nonces
Expiry enforcement	write	Set past deadline
now usage	read	advance time
tx expiry	write	past deadline
schedule	read	claim windows
per-op delay	write	call twice quickly
oracle twap	read	vary window size
delegatecall correct	write	Call via proxy
layout compatibility	offchain	Compare structs/slots
only admin can upgrade	write	Attempt upgrade by non-admin
Upgraded emitted	event	Perform upgrade
upgrade -> rollback	write	Upgrade then rollback
initializer guard	write	call initialize twice
impl selfdestruct	write	Try selfdestruct via impl
read impl/admin	read	Read slots
upgradeToAndCall	write	Upgrade + init call
beacon pattern	write	Swap beacon impl

Key Assertions	Suggested Tools	Notes
Revert for non-owner; succeed for owner	Foundry	
Role gates enforced; events emitted	OpenZeppelin AccessControl	
Cannot execute early; executes after delay	TimelockController	
Only guardian allowed	Custom	
Below threshold fails; threshold passes	Safe CLI + Foundry	
No hijack; only pending can accept	Ownable2Step	
Sig verified; replay blocked	EIP-712	
Access revoked immediately	AccessControl	
Revert	—	
Revert	—	
Blocks action	—	
Access denied	—	
Event emitted; id produced	Bridge mock	
Mint/release executed once	Light client mock	
Refund path works	—	
Second attempt rejected	—	
Queued or revert	—	
Cannot finalize early	Rollup mock	
Gas within cap or revert early	Gas snapshots	
Handled; no funds stuck	Try-catch	
Fails gracefully	Anvil config	
Cost accounted; no overflow	—	
Indexers cope; gas paid	—	
No starvation; pagination	—	
Balances adjust; event emitted	OpenZeppelin tests	
Allowance set; event	—	
Allowance decreases; balances move	—	
Sig valid; nonce inc	EIP-712	
Revert	—	
Reverts on underflow; clamps if custom	—	
Only role can mint/burn	AccessControl	
Static values correct	—	
Events present & correct	—	
Revert when blacklisted/paused	Compliance mod	
Revert on worse price	Uniswap router	
Revert	Risk engine	
Front-running ineffective	Commit scheme	
Invariants hold	Aave mock	
Revert until cooldown	—	
Rate limit trips	—	
Revert	—	
Event emitted with correct args	Foundry logs	
Order stable & documented	Foundry	
No logs on read	Foundry	
Only matching events returned	cast/ethers-rs	
Consumer handles idempotently	Custom indexer	
Indexer does not drop	TheGraph/subgraph	
No timeouts; pagination ok	Batch scans	
Decoder errors handled	Fuzz + decoder	
Votes credited	ERC20Votes	
Only valid states progress	Governor	
>= quorum	—	

Only proposer/guardian	—
Auth via proposal	—
Mitigated by snapshots	—
Decoded types match ABI; no panic on tuple/array	Foundry/Hardhat, ethers.js, alloy/ethers-rs
Contract reads inputs correctly (echo tests)	Foundry cheatcodes
No unintended function executed	Slither/Surya + Foundry
Error selectors & args match	Foundry expectRevert
Topics align with ABI; non-indexed data decodes	Foundry/ethers-rs log decode
Documented rounding held	PRBMath/FixedPoint
Revert or correct behavior	Checked math
Sum of parts == total	—
Not stuck; policy-defined	—
Assets conserved	Invariant test
Bounded error	—
Revert or allowed as spec	—
Matches spec	—
_msgSender correct	Forwarder mock
Rejected by nonce	ERC-4337 bundler mock
UserOp rejected	Paymaster mock
Rejected	Aggregator mock
Handled; no stuck state	EntryPoint
Validation fails	—
Receiver hook passes	ERC721Receiver
Owner updates; event	—
Operator can transfer	—
Correct composition/IPFS	—
Balances updated	—
Hook checks	ERC1155Receiver
Correct receiver/amount	—
Indexes consistent	—
URIs switch correctly	—
Sig grants approval	EIP-4494
>= 95% important writes emit	Coverage tool
Human-readable & unique	—
Docs up to date	solc --userdoc
No leak in prod build	Forge profile
age <= heartbeat	Chainlink mocks
abs% <= limit	—
Require quorum met	Mock oracles
Used when oracle stale	Uniswap math
If down -> revert/failsafe	Arbitrum/Optimism flag
No precision loss	Math lib
Fails safe; no trades	Policy
Clamped/rejected	Fuzz
Revert on write; reads OK	Pausable
Writes succeed	Pausable
Revert for non-pauser	AccessControl
Only affected paths revert	Feature flags
Exceeding rate reverts	TokenBucket
Auto-trip; require manual reset	Circuit module
Return values correct; no side-effects	Foundry, cast call
All results decoded; gas within limits	Multicall, Foundry
Values match archival expectations	Anvil fork, archive RPC

Computed slot returns expected value	Foundry vm.load, cast storage
No revert; returns default	Foundry
Reverts without role; passes with role	Foundry roles
Reverts on writes under staticcall	Foundry
No unexpected drift across 3 blocks	WS provider
No reentrancy effect	Reentrant tester
Reverts on reenter	Guard
No double-withdraw	—
Guard holds	ERC777 mock
Guard holds	Custom
Invalid signature	Foundry/ethers
Replay blocked	ERC20Permit
Revert	—
Revert; events	Permit2
Rejected	secp256k1 rules
Fail gracefully	—
State changed; event emitted	Foundry
All side effects occur atomically	Foundry/Router
Single effect; no duplication	Foundry/ethers-rs NonceManager
Replay fails	Anvil forks
Gas reduced; no behavioral change	geth/Anvil
Correct custom errors emitted	expectRevert
Tx rejected; error code	Anvil config
Allowance set then spent	ERC20Permit
Mempool handling; eventual success	Local node
Revert with Expired	Foundry
Time math correct	vm.warp
Revert	—
Claims limited to schedule	—
Second call blocked	—
Window honored	—
State changes in proxy storage	UUPS/Transparent
No slot collision; gaps kept	Storage layout checker
Revert unauthorized	UUPS/Transparent
Event contains new impl	Event decode
State consistent; no bricking	Playbook test
Second call reverts	Initializable
Proxy safe; no bricking	Hardhat/Foundry fork
Addresses match expectations	cast storage
Both succeed atomically	UUPS
All proxies point to new impl	Beacon